

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
Work Integrated Learning Programmes Division
Cluster Programme - M. Tech in AI & ML
Ist Semester , 2025 – 26(Jan,2026)
COMPREHENSIVE Examination (Regular)

Course No. : AIMLZG511
Course Title : Deep Neural Network
Nature of Exam : Open Book
Weightage : 40%
Duration : 2.5 Hours
Date of Exam : 23/05/2026 (FN)

No. of Pages	= 02
No. of Questions	= 06

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.
4. Use of Scientific Calculator is permitted.

Q1) You are an ML engineer debugging a deep learning model that is failing during training.
System details: Input $x = 1$, Each layer has a single scalar weight, All weights initialized to $w = 2$,
 $dL/dy = 1$, Learning rate $\eta = 0.01$ **(02+01+02+03=08Marks)**

Observed logs:

Epoch 1: Loss = 0.8, Gradients approx 10–20
Epoch 3: Loss = 15.2, Gradients approx 300+
Epoch 5: Loss = NaN

- (a) Based on the logs, estimate whether the model is shallow (around 5 layers) or deep (around 10 layers). Support your answer with a quick calculation.
- (b) Compute the gradient with respect to the first weight for a 10-layer network.
- (c) Diagnose the issue: (i) What is the root cause of the failure? (ii) Why does the loss become NaN?
- (d) Propose practical fixes: Explain each briefly.
 - (i) One initialization fix
 - (ii) One training-time fix
 - (iii) One architecture-level fix

Q2) A CNN trained for defect detection performs well on Camera A but fails on Camera B after deployment.

Observed logs: Training accuracy: 96% **(01+02+02+02=07 Marks)**
Camera A accuracy: 94%
Camera B accuracy: 58%
Camera B images: lower illumination, motion blur, different angle

- (a) Identify the most likely root cause. Is this overfitting, underfitting, or domain shift? Justify using evidence.
- (b) Design a step-by-step debugging plan (at least 3 steps) to isolate the exact failure point in the pipeline.
- (c) Propose one preprocessing or data-centric fix and explain why it directly addresses the issue.
- (d) If accuracy must improve without retraining the full model, what practical workaround would you deploy?

Q3) Encoder hidden states: $h_1 = 0.3, h_2 = 0.5, h_3 = 0.2$ (01+01+02+02+02=08 Marks)

Attention weights: $\alpha_1 = 0.2, \alpha_2 = 0.6, \alpha_3 = 0.2$

- Compute the context vector c_t .
- If h_2 is perturbed to 0.1 due to noise, recompute c_t and quantify the change.
- Based on (i) and (ii), analyze sensitivity of attention to dominant weights.
- In a long sequence setting, explain one limitation of attention weights like these.
- What is exposure bias and how could it interact with attention errors during inference?

Q4) A transformer has the following values,
 $Q = [2, 0], K = [2, 2], V = [1, 3], dk = 2$

(03+01+02=06 Marks)

- (i) Compute dot-product attention score (without scaling and without softmax).
 - Compute scaled score.
 - Explain how scaling changes the magnitude and why this matters during softmax.
- (b)** During training of a Transformer with $dk = 512$, attention scores before softmax become very large (greater than 100), the softmax outputs are nearly one-hot, and gradients are close to zero. It is later found that the scaling factor $1/\sqrt{dk}$ was removed. Explain why this happens.
- (c)** You are designing a system with multi-head attention. Instead of increasing the number of heads, you increase the embedding dimension.
- What trade-off does this introduce?
 - Explain how using multiple attention heads changes the type of representations learned compared to using a single high-dimensional attention mechanism.

Q5) Given a grayscale image:

(01+01+01+01+02=06 Marks)

Input (3×3)	Filter 1 (3×3)	Filter 2 (3×3)
[1 2 1]	[1 0 -1]	[0 1 0]
[0 1 0]	[1 0 -1]	[1 -4 1]
[2 1 2]	[1 0 -1]	[0 1 0]

Stride = 1, Padding = 1

- Compute the output feature maps for both filters. Apply ReLU
- Which filter is detecting edges more strongly? Justify using output values.
- If both filters produce very similar outputs on a large dataset, what does it imply about filter diversity? What action would you take?
- During deployment, you notice that outputs from Filter 1 are consistently higher on one side of the image. What does this indicate about the type and direction of edges present in the data?
- If you deploy this model on higher-resolution images (e.g., 256×256), what issue might arise with using the same filters? Suggest one improvement.

Q6) A practitioner switches from SGD to Momentum using: $\beta=0.95$. The training loss oscillates more than before. Given gradients: (02+01+02=05 Marks)

Iteration	Gradient
1	12
2	11
3	10

- Compute velocity after iteration 3 assuming initial velocity is zero.
- Explain why accumulated velocity can overshoot minima.
- Suggest two hyperparameter fixes.